

TT-Bio at the booth

An AI predicts the 3D shape of a protein from its sequence, live, on the four Tenstorrent Blackhole chips inside this QuietBox. The screen folds one protein after another by itself. Nobody needs to log in or touch the box once it is on.

Set up

1. **Place the box** with its vents clear. Do not open it.
2. **Screen:** HDMI cable from the box's **motherboard panel** (the HDMI port among the USB ports) to the screen. The Tenstorrent cards have no display output; plug nothing into their brackets. Switch the screen on and set its input to that HDMI. **Connect the screen before powering on.**
3. **Keyboard:** a USB keyboard in any USB port. Visitors do not type for now; it is for Tab and Esc. A touch screen also needs its USB cable.
4. **Network (optional):** booth Ethernet into the box's network port. The demo never uses it; it only lets us help remotely.
5. **Power:** plug in, press the power button once.

What you should see

after	on the screen
under 1 min	a still picture of a folded protein
~1¼ min	points condensing into a protein, labelled Recorded folds ; the four chip rows say <i>warming up</i>
~1½ min	the label turns Live on four Blackhole chips and each chip row names what it is folding

A chip row that says *recovering* or *resetting* for a few minutes is the box repairing that chip while the others carry on; leave it. Typing a name to fold it is switched off for now (?visitors=1 on the kiosk URL brings it back). **Tab** shows the four chips' power, temperature and what each is folding, Tab again returns. After a minute without input the screen goes back to the loop by itself.

If something goes wrong

you see	do this
No picture	Check the cable sits in the motherboard's HDMI and the screen's input matches. If the screen was connected after the box was switched on and nothing appears within 10 seconds, restart it (below).
A frozen or black screen	Wait. The box repairs itself, usually within a minute. Rarely the whole box freezes; then it restarts itself within about 3 minutes and is folding again within 10. If the screen is still frozen or black after 10 minutes, restart it.
Recorded folds for more than 10 minutes, or every visitor gets <i>The chips are busy</i>	Leave it running: what it shows is real, recorded on this box, and labelled as such. Call.
The demo will not come back after a restart	Play the video: Ctrl+Alt+F3 , log in, ~/tt-bio-booth/demo/booth/ops/boothctl video, Ctrl+Alt+F2 .

Restart: press the power button once, briefly. The box stops the demo cleanly (this can take up to 2 minutes) and is back on it about 1½ minutes later. Do not hold the button: holding it for 5 seconds cuts power, and then it stays off until someone presses it again.

End of day: leave the box running and switch only the screen off. If the hall cuts power overnight, press the power button in the morning.

Who to call

Moritz Thüning, phone _____, mthuenig@tenstorrent.com. Say what the screen shows and since when. With the Ethernet in, Moritz can fix most things remotely.

For whoever answers the call: ~/tt-bio-booth/demo/booth/ops/boothctl health, as the user who installed the demo, answers HEALTHY, REPAIRING ITSELF or NOT HEALTHY in plain words, one line per part; read it out as it is. boothctl status | restart |

logs for the rest. A restart can take 90 s or more to stop the engine; never send it SIGKILL. Details in `ops/README.md`; talking points in `TALKING.md`.